

TAWERS[®] WG4

High deposition enables
high-speed TIG welding

TAWERS-TIG

High-frequency start



TAWERS-TIG start unit

Achieves excellent arc
start. Enables improved
welding quality and
reduces takt time.



Not applicable to aluminum.

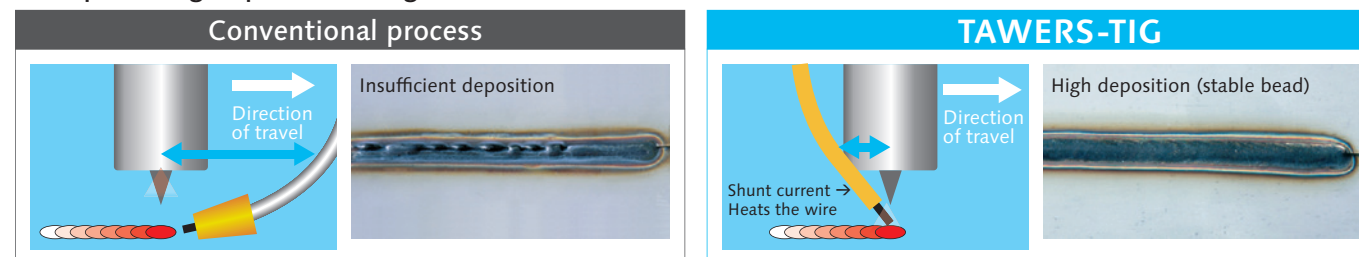
WG4

TS	TM	TL	LA
800	1100	1800	1800
950	1400		

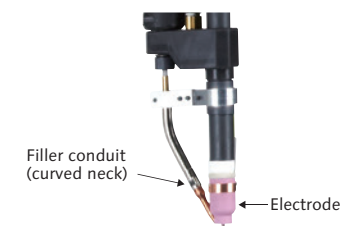
* TS: External
* TM: External
* TL: External
* LA: External

The proximity of the electrode and filler wire increases
the wire heating effect. Example of high-speed welding
(80 cm/min, stainless steel)

Example of high-speed welding (80 cm/min, stainless steel)

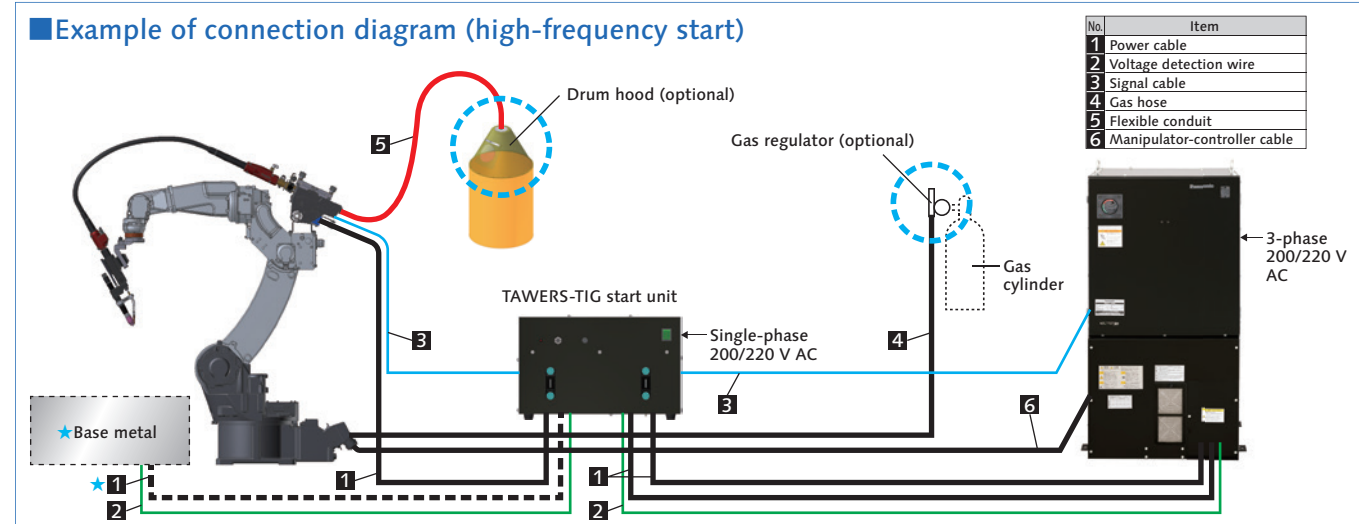


Curved neck filler conduit



Achieves consistent filler wire feeding.
Effective in improving weld quality and limiting misalignment.

Example of connection diagram (high-frequency start)



★ Items to be supplied by the customer

Please contact us for details.

Robot Systems G4

Realizes stable high-quality
welding in combination
with a full digital welding
power source



Full Digital CO₂/MAG Welding Machine NE1 Series **SP-MAG** SP(Super-imposition Control)

CO₂/MAG/MIG welding robot system that can be selected according to your application

TS	TM	TL	LA
800	1100	1800	2000
950	1400	2000	
	1600		
	1800		
	2000		



400NE1

Adopts SP control, which has been
installed in the world's first welding
power source integrated robot TAWERS,
and has been praised by many customers.

Features of SP-MAG

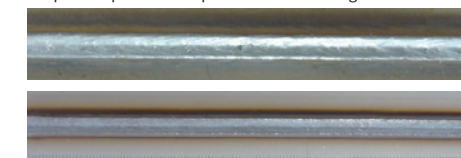
- Spatter reduction (reduced man-hours for removal)
- Optimal for high-speed welding due to a shortened short-circuit cycle
- Beautiful bead appearance achieved by shortened arc length



TM-1400G4 (Separate)

Beautiful bead appearance and reduced spatter achieved even in high-speed welding by GZ4

* Optional parts are required for connecting a robot.



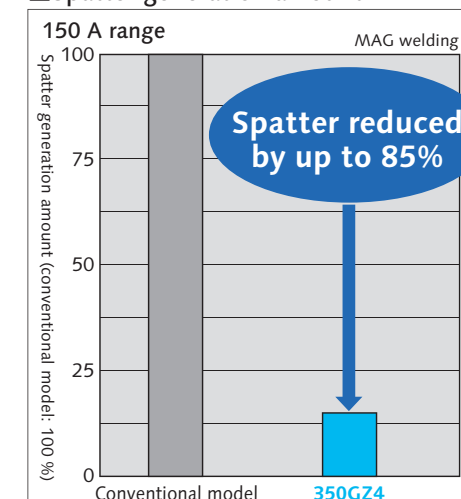
MAG welding (220 A)

Joint: Fillet Base metal: Mild steel SPCC (thickness: 2.3 mm) Welding current: 220 A
Welding speed: 100 cm/min Wire: ϕ 1.2 (YM-50MT) Gas: MAG (80% Ar and 20% CO₂)

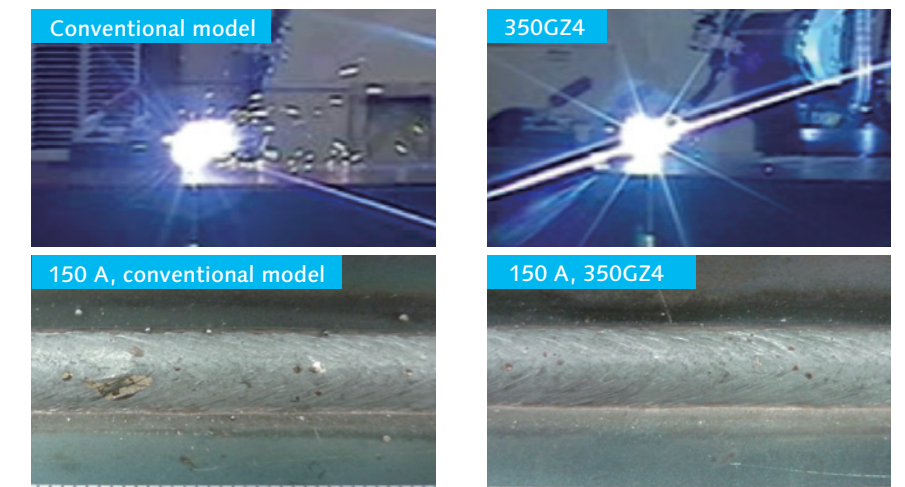
MIG welding (180 A)

Joint: Fillet Base metal: SUS308 (thickness: 1.5 mm) Welding current: 180 A
Welding speed: 80 cm/min Wire: ϕ 1.2 (Y308LSi) Gas: MIG (98% Ar and 2% O₂)

Spatter generation amount



* Up to 80% reduction in 500GZ4
(Compared to conventional model, 250 A range)



Joint: Fillet Base metal: Mild steel SPCC (thickness: 2.3 mm) Welding current: 150 A
Welding speed: 50 cm/min Wire: ϕ 1.2 (YM-50MT) Gas: MAG (80% Ar and 20% CO₂)

Full Digital Controlled Welding Machine **Full Digital**

Lineup of CO₂/MAG/
MIG welding machines
to achieve high-quality
welding



400NE1

350GZ4

500GZ4

350VZ1TA1

350VR1TA1

500VR1TA1

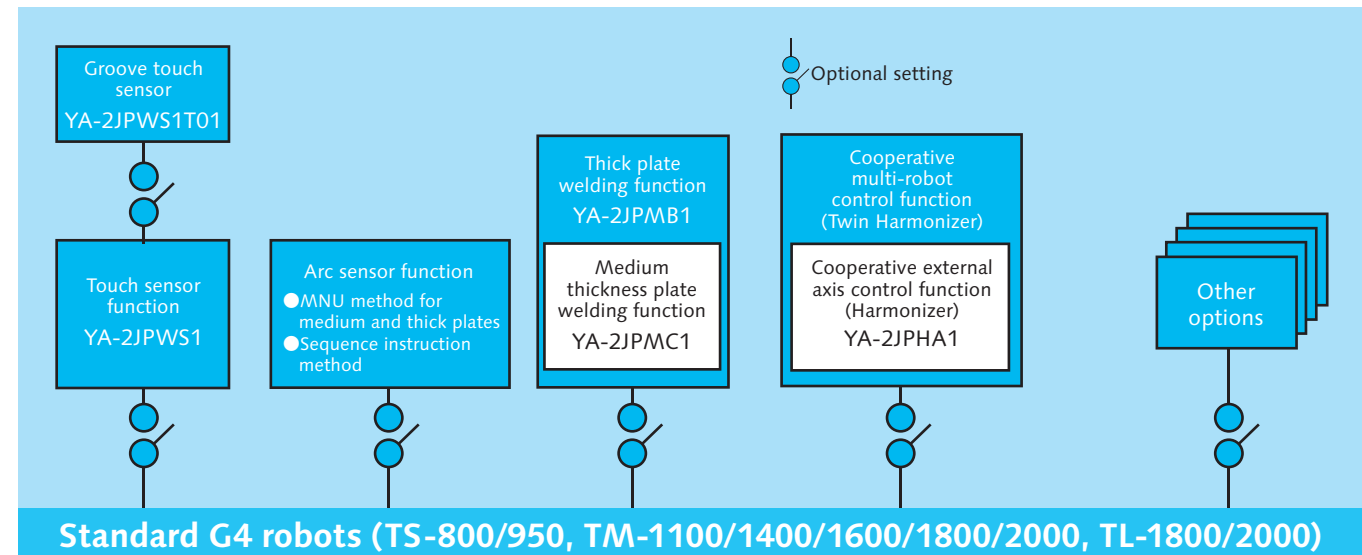
400VP1TA1

Medium and Thick Plate Welding Robot System G4

Freely selectable functions effective for medium and thick plate welding

Medium and thick plate welding system

Functions effective for medium and thick plate welding



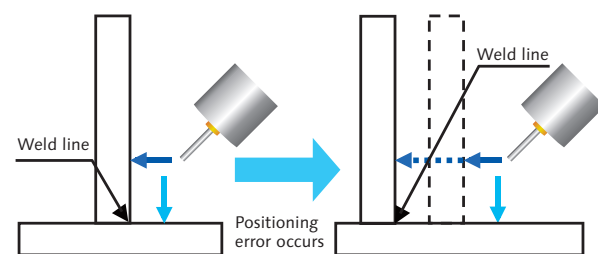
* Please contact us for details.

Examples of functions

Touch sensor operation

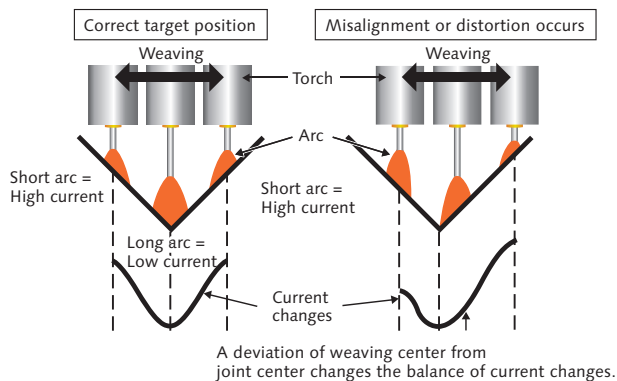
The robot calculates the point of contact with the base metal and determines the weld line.

Measures deviation in the weld line due to errors and determines the weld line again.



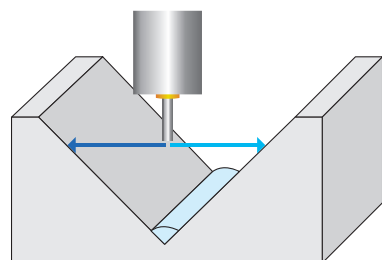
Arc sensor operation

Detects misalignment or distortion of the workpiece and adjusts the position to the correct target position.



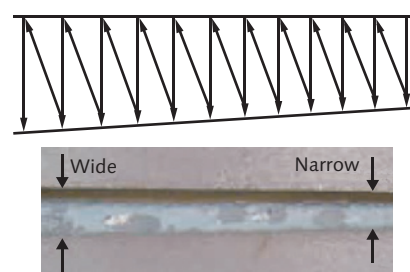
Groove touch sensor function

Searches for the groove and detects positioning errors. Senses groove width and center, compensating for misalignment in each workpiece.



Variable weaving function Thick plate welding function (YA-2JPMB1)

Adapts to changes in the groove width. Controls the amount of deposited metal, ensuring uniform bead height.



TIG Robot System G4

Realizes high quality welding in combination with a full digital welding power source

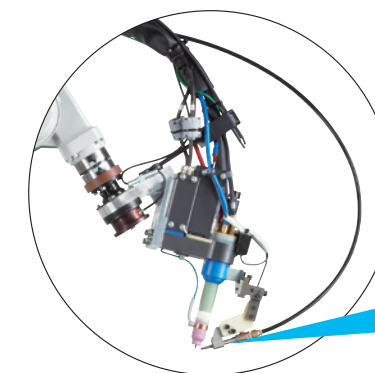
TIG welding robot system that can be selected according to your application

Combinations of applicable materials, welding power sources, and robots

Type	Material	Applicable filler wire diameter (mm)	Applicable welding power source	Applicable robot
TIG without filler	Stainless steel	—	300BZ3	TS-800 TS-950 TM-1100 TM-1400 TL-1800 LA-1800
	Stainless steel Aluminum	—	300BP4 500BP4	
TIG with filler	Stainless steel	1.2	300BZ3	TS-800 TS-950 TM-1100 TM-1400 TL-1800 LA-1800
	Stainless steel Aluminum	1.2	300BP4 500BP4	
Rotary TIG with filler	Stainless steel	1.2	300BZ3	TL-1800 LA-1800
	Stainless steel Aluminum	1.2	300BP4 500BP4	

* An external axis controller is required for the rotary TIG filler welding.

Features of the rotary TIG filler unit



- Optimal welding position achieved
- High-precision filler feeding
- Improved accessibility to workpieces

Filler tip position can be adjusted up/down, right/left, and front/back

Lineup of TIG welding torches



Full Digital Controlled Welding Machine

Full Digital

Lineup of TIG welding machines to achieve high-quality welding

AC/DC TIG welding machines



300BP4



500BP4

DC TIG welding machine



300BZ3